ENTMLGY 6707 Entomological Techniques and Data Analysis

Multivariate community analyses

Characterizing insect communities with multivariate methods

Assessing patterns of species diversity is a common approach used in ecology to understand communities. However, the use of diversity indices has its limitations.

Although these indices provide information about the number of species at different sites, this measure doesn't provide any indication as to whether those species are the same or not. For example, two insect communities may each have 10 species. The calculation for species richness indicates that these two communities are similar in terms of their diversity. What we don't know is whether the same 10 species are shared between these two communities. How similar is the overlap in species composition between these two sites? Species diversity indices measured at the site-level (i.e., alpha-diversity) are often paired with ordination methods that can assess measures of beta-diversity, or the change in species composition among sites.

Diversity metrics take data from multiple species collected at a site and distill this information into a single value. The calculated diversity metric can then be used as a response variable in a univariate model (i.e., GLM or GLMM). Multivariate models allow for the analysis of multiple environmental factors on many species simultaneously by representing relationships among variables in a low-dimensional space. These approaches are also capable of handling noisy and redundant data, as well as sparse data (i.e., many zeros from species that occur at low frequencies). This is convenient because community data tend to contain more than one response variable, and it is a challenge to understand the complexity of species responses using univariate approaches. Multivariate techniques are able to represent data along a reduced number of orthogonal axes (i.e., linearly independent, uncorrelated) such that they represent (in decreasing order) the main trends of the data.

Multivariate analyses aid in addressing some of these limitations, and therefore, can provide a more holistic understanding of community responses. These methods can reveal patterns and relationships in data sets by reducing the dimensions to a more manageable form. There are two main classes of multivariate community analyses: classification and ordination. Classification centers on the placement of species and/or samples into groups, while ordination focuses on the arrangement or order of species and/or samples along gradients. To some degree, these approaches can be viewed as complementary. Both approaches require a species by sample matrix. We are going to focus on ordination approaches.

We will use two open source data sets to demonstrate how to run and interpret several common multivariate analyses. These data are ant and understory plant species collected in the Hemlock Removal Experiment at the NSF Harvard Forest Long-term Ecological Research (LTER) site (See more here: https://harvardforest1.fas.harvard.edu/exist/apps/datasets/showData.html?id=HF118). This experiment includes four treatments (Hemlock girdled, Hemlock logged, Hemlock control, and Hardwood control) each replicated across two ($n = 2$) 90 x 90 m plots. The goal of this experiment was to investigate the effects of hemlock mortality from the invasive hemlock woolly adelgid and preemptive hemlock removal via logging. The specific study that we are using focuses on ant community responses to these environmental changes, but there have been many studies conducted within this experiment.

Because the data are openly available on the NSF LTER website, we can directly load them into R.

```
ants <- read.csv(file="https://pasta.lternet.edu/package/data/eml/knb-lter-hfr/118/35/90ca76
            header=T, na.strings=c("",".","NA"))
```

Notice that the ant data are in `long format`. Each row contains the count of one species at a specific site and date. Ant species are combined into one column. To run the multivariate analyses, we want the data represented in `wide format`. In this case, each species and their associated counts will be represented in a separate column. Each row will represent the counts of all ant species at a specific site and date. We can reshape the data set using the `reshape2` package.

```
ant.matrix <- dcast(ants, year + block + plot + treatment + trap.type ~ code,
                    sum, value.var = "abundance", na.rm = TRUE)

# change variables to factors
ant.matrix$year <- as.factor(ant.matrix$year)
ant.matrix$block <- as.factor(ant.matrix$block)
ant.matrix$plot <- as.factor(ant.matrix$plot)
ant.matrix$treatment <- as.factor(ant.matrix$treatment)
ant.matrix$trap.type <- as.factor(ant.matrix$trap.type)

levels(ant.matrix$year)
```

```
 [1] "2003" "2004" "2005" "2006" "2007" "2008" "2009" "2010" "2011" "2012"
[11] "2013" "2014" "2015" "2018"
```

The data set includes ant species collected over multiple years. Let's subset the data to focus on ants collected via pitfall traps in 2015 and 2018. Since it is likely that some of the ant species represented in the full data set (2003-2018) were not collected in these years (especially rarer species), let's also remove the columns of species that were not collected.

```
yr1 <- ant.matrix %>% filter(year == "2015") %>% droplevels()
yr2 <- ant.matrix %>% filter(year == "2018") %>% droplevels()

ants2 <- rbind(yr1, yr2)

ants3 <- ants2[, colSums(ants2 !=0) > 0]
```

Now let's load the understory plant data. This data set is also openly available on the NSF LTER website.

```
herb <- read.csv(file="https://pasta.lternet.edu/package/data/eml/knb-lter-hfr/106/33/417bcd2
            header=T, na.strings=c("",".","NA"))
```

We need to reshape the data into `wide format`. Let's subset the shrub and herbaceous data so that we are working with the same years - 2015 and 2018. We will also remove plant species that are not very common in the data set.

```
herb$species <- as.factor(herb$species)
# species codes are provided in the csv file hf106-01-species-codes

herb.matrix <- dcast(herb, year + block + trt + plot ~ species,
                     mean, value.var = "cover", fill = 0)

yr1.herb <- herb.matrix %>% filter(year == "2015") %>% droplevels()
yr2.herb <- herb.matrix %>% filter(year == "2018") %>% droplevels()

herb2 <- rbind(yr1.herb, yr2.herb)

# remove columns of species that were not observed or observed at low frequencies
herb3 <- herb2[, colSums(herb2 !=0) > 2]
```

The ant and plant data sets are ready to go. We will use several multivariate methods to assess ant community response to the hemlock treatments. Percentage cover of understory plant species will be used as additional predictor variables to explain ant communities in these sites.

Ordination methods

There are two types of ordination techniques that differ based on the incorporation of environmental data: indirect and direct gradient analysis. We are able to run both types of these ordination analyses using the R package `vegan`.

Indirect gradient analysis

Indirect gradient analysis (aka unconstrained ordination) utilizes only the species by sample matrix. Any environmental data are used after the analysis to aid with the interpretation. When using an indirect gradient analysis, the approach identifies gradients solely based on the species data. Therefore, knowledge of the species and system are important for interpreting the results. Two examples of indirect gradient analysis are Nonmetric Multidimensional Scaling (NMDS) and Principal Component Analysis (PCA).

Nonmetric multidimensional scaling (NMDS)

Nonmetric multidimensional scaling is a commonly used technique to assess differences in species composition among samples, sites, treatments, etc. It is an iterative method that maximizes the rank order correlations between the distances in the dissimilarity matrix and the distances in low-dimensional space such that the output represents (as well as possible) the ordering among sites in species space.

NMDS is a distance-based technique that relies on a dissimilarity matrix created from incidence-based (i.e., presence/absence) or abundance-based species data. Dissimilarity matrices are essentially a measure of beta-diversity, which is the diversity `between` samples, sites, treatments, etc. The species diversity indices that we have already discussed are all measures of alpha-diversity, which is the diversity `within` a particular sample, site, treatment, etc. Gamma-diversity represents the total species diversity across all samples, sites, treatments, etc. Alpha-diversity and gamma-diversity are partitions of diversity at different scales, and beta-diversity describes the link or change between these two scales. With measures of beta-diversity, we can ask questions about how diversity changes among sites and evaluate whether sites are similar (or not) in terms of species diversity.

There are many different beta-diversity metrics, all of which have slightly different equations. The appropriate beta-diversity metric depends on the study question, data availability, and the ecology of the species. Because species richness tends to increase with sample size, beta-diversity is calculated between pairs of samples. A larger beta-diversity value means higher dissimilarity (i.e., greater difference) among those two communities in terms of their species composition. A lower beta-diversity value means lower dissimilarity among communities (i.e., the communities are more similar in terms of their species composition). A dissimilarity matrix can be calculated using the function vegdist(). The dissimilarity metric to be calculated is indicated by the `method` argument.

```
# change ant data to presence/absence
ants4 <- ants3[,6:38] # save as a new object
ants4[ants4 > 0] <- 1 # if a value is greater than zero, replace it with 1
ants4$treatment <- ants3$treatment # add the treatments back into the data set

dis.matrix.pa <- vegdist(ants4[,1:33], method = "jaccard")
```

You can run a NMDS model in R using the metaMDS() function in the R package `vegan`. The algorithm is iterative. It starts from an initial distribution of samples in the ordination space (often this initial distribution is random), and then each iteration reshuffles the samples in search of the optimal positions of `n` samples in `k` dimensional space. Each iteration results in a slightly different solution, but the aim is to identify the best solution that minimizes the stress of the configuration. `Stress` is a measure of the mismatch between the rank order of the dissimilarities and the low dimensional solution. The default is k = 2 (i.e., 2 dimensional). Lower stress is better and ideally the stress should be below 0.2. Essentially, the stress is a measure of the fit of the model and should be reported in the results.

```
nmds.ants.pa <- metaMDS(dis.matrix.pa, trymax = 500, autotransform = TRUE, k = 2)
```

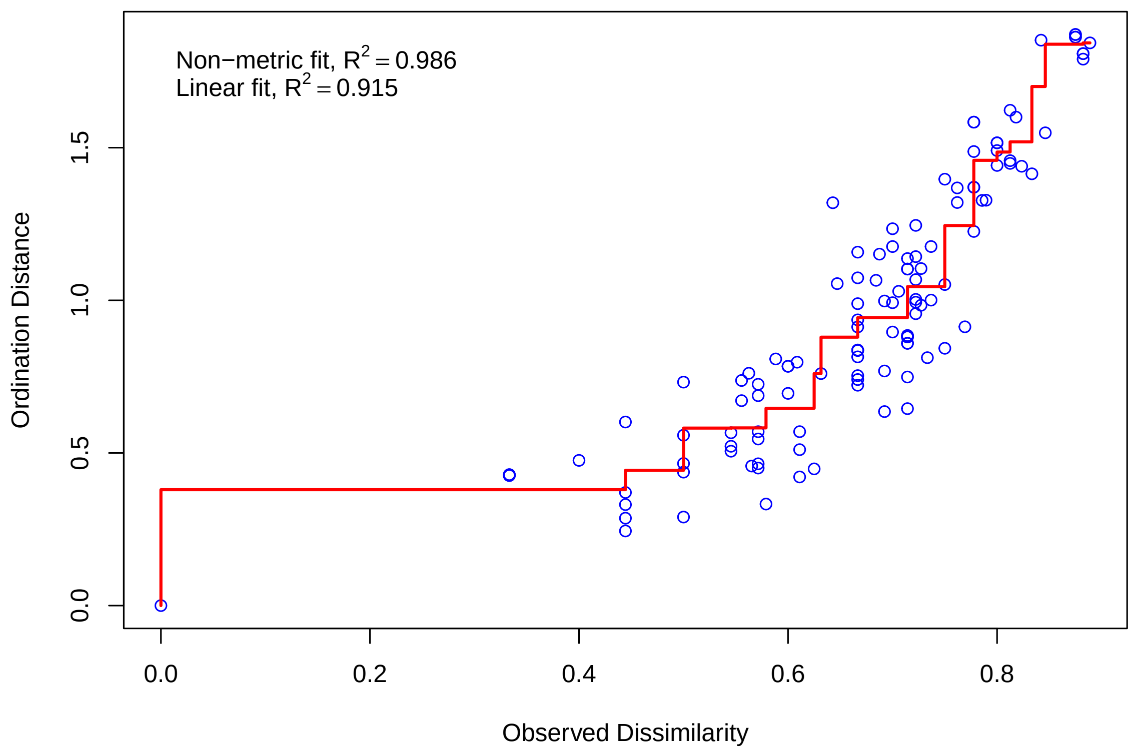
```
Run 0 stress 0.1183155
Run 1 stress 0.1183155
... Procrustes: rmse 7.569497e-05  max resid 0.0001415854
... Similar to previous best
Run 2 stress 0.1848854
Run 3 stress 0.1183155
... Procrustes: rmse 4.091166e-05  max resid 7.745098e-05
... Similar to previous best
Run 4 stress 0.2791146
Run 5 stress 0.1496685
Run 6 stress 0.1493727
Run 7 stress 0.1667825
Run 8 stress 0.1183155
... Procrustes: rmse 1.776435e-05  max resid 3.270022e-05
... Similar to previous best
Run 9 stress 0.1637824
Run 10 stress 0.1183155
... Procrustes: rmse 5.528733e-05  max resid 0.0001025303
... Similar to previous best
Run 11 stress 0.1651467
```

```
Run 12 stress 0.1305361
Run 13 stress 0.1545166
Run 14 stress 0.1496685
Run 15 stress 0.1183155
... Procrustes: rmse 5.815556e-05  max resid 0.0001076828
... Similar to previous best
Run 16 stress 0.1305361
Run 17 stress 0.1305361
Run 18 stress 0.1183155
... Procrustes: rmse 0.0001036427  max resid 0.0001964842
... Similar to previous best
Run 19 stress 0.1765853
Run 20 stress 0.1667825
*** Best solution repeated 6 times
```

```
nmds.ants.pa
```

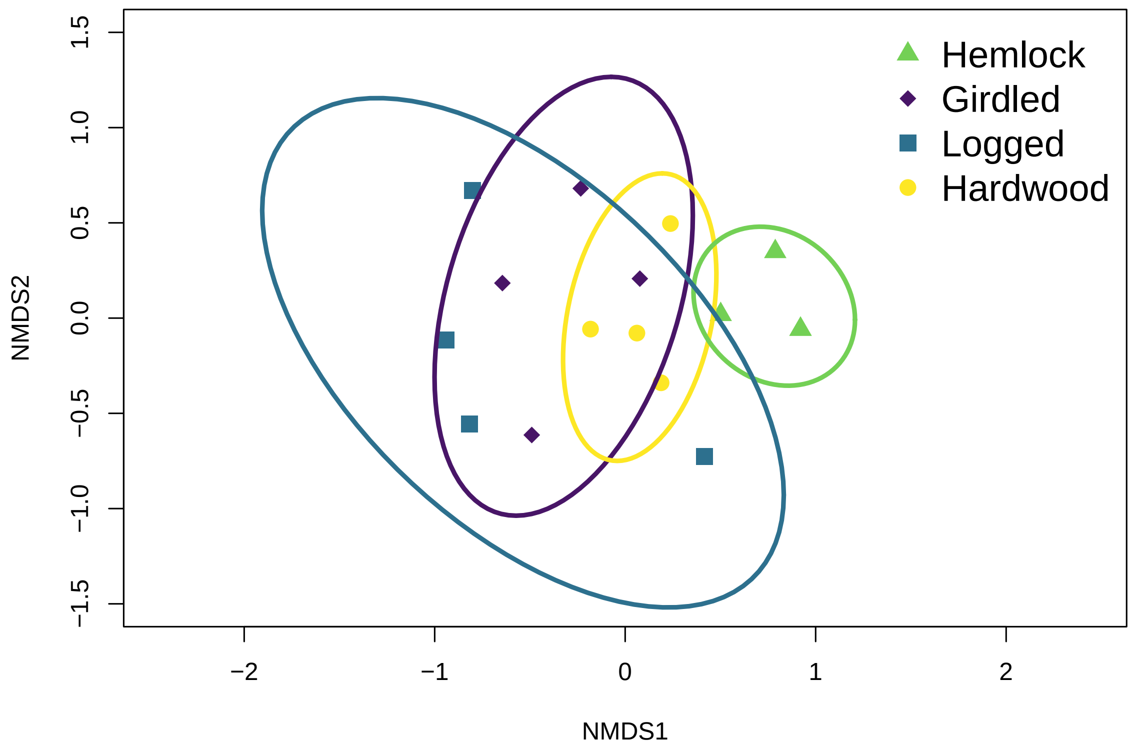
```
Call:
metaMDS(comm = dis.matrix.pa, k = 2, trymax = 500, autotransform = TRUE)

global Multidimensional Scaling using monoMDS

Data:     dis.matrix.pa
Distance: jaccard

Dimensions: 2
Stress:     0.1183155
Stress type 1, weak ties
Best solution was repeated 6 times in 20 tries
The best solution was from try 0 (metric scaling or null solution)
Scaling: centring, PC rotation, halfchange scaling
Species: scores missing
```

```r
stressplot(nmds.ants.pa)
```




```r
ordiplot(nmds.ants.pa, disp = "sites", type = "n", xlim = c(-1.5, 1.5),
         ylim = c(-1.5, 1.5))
points(nmds.ants.pa, dis = "sites", select = which(ants3$treatment=="HemlockControl"),
       pch = 17, cex = 1.5, col = "#73D055FF")
points(nmds.ants.pa, dis = "sites", select = which(ants3$treatment=="Girdled"),
       pch = 18, cex = 1.5, col = "#481567FF")
points(nmds.ants.pa, dis = "sites", select = which(ants3$treatment=="Logged"),
       pch = 15, cex = 1.5, col = "#2D708EFF")
points(nmds.ants.pa, dis = "sites", select = which(ants3$treatment=="HardwoodControl"),
       pch = 16, cex = 1.5, col = "#FDE725FF")
ordiellipse(nmds.ants.pa, ants3$treatment, draw = "lines",
            col = c("#481567FF", "#FDE725FF", "#73D055FF", "#2D708EFF"),
            lwd = 3, kind = "sd", conf = 0.90, label = FALSE)
```

```
legend("topright", legend = c("Hemlock", "Girdled", "Logged", "Hardwood"),
       pch = c(17, 18, 15, 16), cex = 1.5, bty = "n",
       col = c("#73D055FF", "#481567FF", "#2D708EFF", "#FDE725FF"))
```



The NMDS plot shows the distribution of sites in low-dimensional space, with the sites coded based on hemlock treatment. This plot is a visual representation of the NMDS model. The stress was 0.11, which, along with a high R2 value from the Shepard plot, indicate that the model is a good fit for the data. The R2 in the Shepard plot is determined by comparing the relationship between the final distances of the objects in the ordination and the original distances of the community data. Other than these measures of fit, there are no statistical results.

NMDS is a distance-based technique (as opposed to an eigenvector-based technique which is discussed below), so it does not maximize the variability associated with the individual ordination axes. Therefore, there is no measure of the variability explained by the axes. In fact, the direction and numeric scale of the axes are not important for the interpretation.

Because the NMDS model is a visual representation of the differences in species composition among sites, this method is often paired with permutational multivariate analysis of variance (PERMANOVA) and homogeneity of multivariate group dispersion (BETADISPER).

Permutational multivariate analysis of variance (PERMANOVA)

PERMANOVA tests whether the group centroid of the communities differs among a categorical grouping factor (e.g., treatments) in multivariate space. The centroids are calculated for each group and the sums of squares of deviations are determined for these points. Significance tests are conducted using permutations of the community data and compared to the sums of squares that were calculated for the observed data.

```
adonis2(dis.matrix.pa ~ ants4$treatment, permutations = 999)
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = dis.matrix.pa ~ ants4$treatment, permutations = 999)
         Df SumOfSqs      R2      F Pr(>F)
Model     3   1.4433 0.41295 2.8137  0.002 **
Residual 12   2.0518 0.58705
Total    15   3.4950 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
pairwise.adonis(dis.matrix.pa, ants4$treatment)
```

```
                              pairs Df SumsOfSqs   F.Model        R2 p.value
1                 Logged vs Girdled  1 0.2406823 0.9936814 0.1420827   0.472
2          Logged vs HemlockControl  1 0.8663977 5.4361524 0.4753480   0.033
3         Logged vs HardwoodControl  1 0.3708324 1.8426037 0.2349480   0.056
4         Girdled vs HemlockControl  1 0.6411288 4.5565102 0.4316304   0.027
5        Girdled vs HardwoodControl  1 0.2146477 1.1756126 0.1638345   0.395
6 HemlockControl vs HardwoodControl  1 0.5528551 5.5425213 0.4801829   0.029
  p.adjusted sig
1      1.000
2      0.198
3      0.336
4      0.162
5      1.000
6      0.174
```

Homogeneity of multivariate group dispersion (BETADISPER)

BETADISPER tests whether the dispersion of a categorical grouping factor from its spatial median is different between groups. This test is essentially a multivariate analogue of Levene's test for homogeneity of variances. Significance is determined using a permutation procedure.

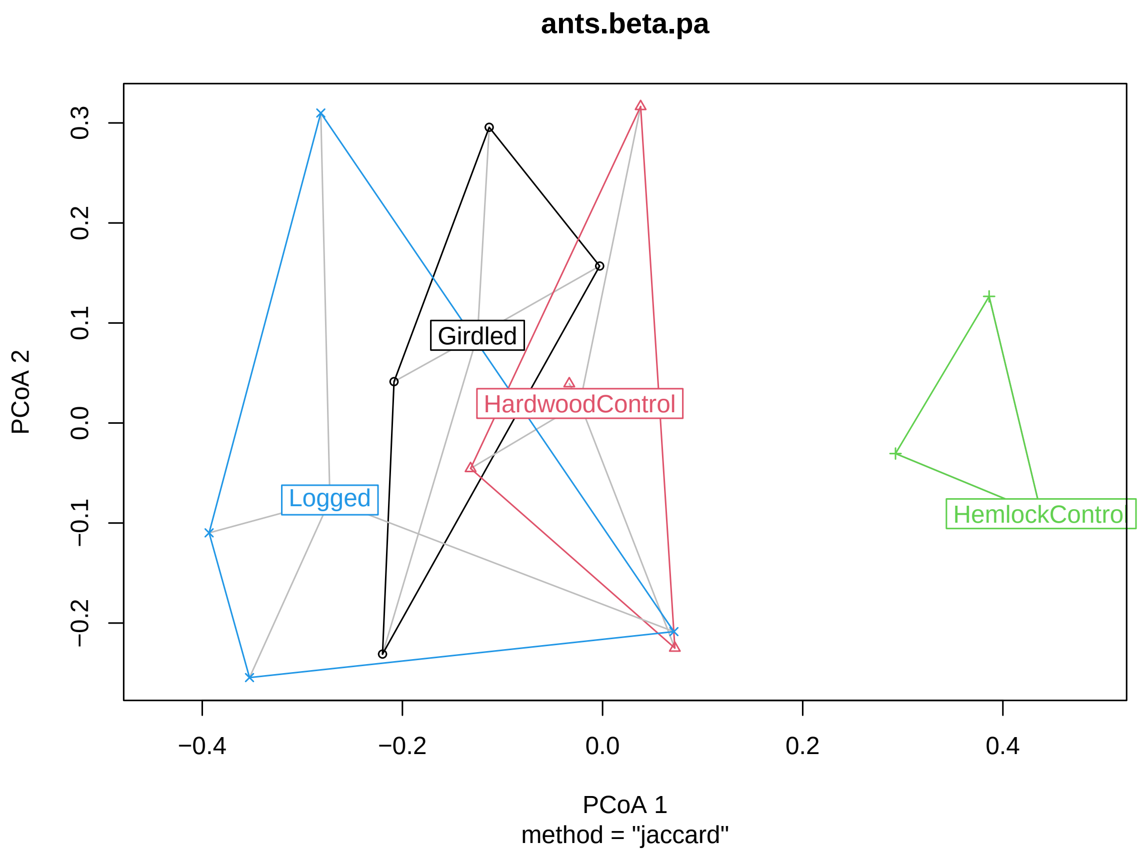
```r
ants.beta.pa <- betadisper(dis.matrix.pa, ants4$treatment, type = c("median"))
```

```
Warning in betadisper(dis.matrix.pa, ants4$treatment, type = c("median")): some
squared distances are negative and changed to zero
```

```r
anova(ants.beta.pa)
```

```
Analysis of Variance Table

Response: Distances
          Df  Sum Sq  Mean Sq F value  Pr(>F)
Groups     3 0.17539 0.058465  3.7923 0.04009 *
Residuals 12 0.18500 0.015417
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
plot(ants.beta.pa)
```



Principal component analysis (PCA)

Principal component analysis is an eigenvalue-based ordination method that uses Euclidean distances among samples to maximize the linear correlation between the distances in the dissimilarity matrix and the distances in low dimensional space. This method is not a `statistical test`, but rather describes the correlation structure among different variables.

You can run a PCA using the function rda() in the package `vegan`. Each ordination axis is an eigenvector with an associated eigenvalue, which represents the variance explained in the model. Axes are ranked by their eigenvalues such that the order of axes is important. Therefore, the first axis has the highest eigenvalue and explains the most variation in the data. Based on the output of the model, the eigenvalues are examined for their importance (i.e., variance explained), and then the patterns of data represented by the axes are described. Report the percentage variance explained by the first two axes.

The linear nature of PCA often results in the arch effect (aka the horseshoe effect), which occurs when the data are distorted in the ordination plot producing a semicircular pattern.

This makes visual interpretation of the patterns difficult. Species data are especially susceptible to the arch effect. Because of this, PCA is mostly used to characterize environmental variables among sites. For example, sample or site scores on PCA axes can be used as predictor variables in other analyses (e.g., GLMs), or environmental variables highly correlated to PCA axes can be selected for further analyses.

We will run a PCA using the herbaceous understory plant data. If the data set contains environmental variables that have different units, these should be standardized to have a mean of 0 and unit standard deviation, Otherwise the variable with the higher absolute value or variance will be more important in the analysis. Since all the plant data are percentage cover, we do not need to standardize (i.e., scale = FALSE).

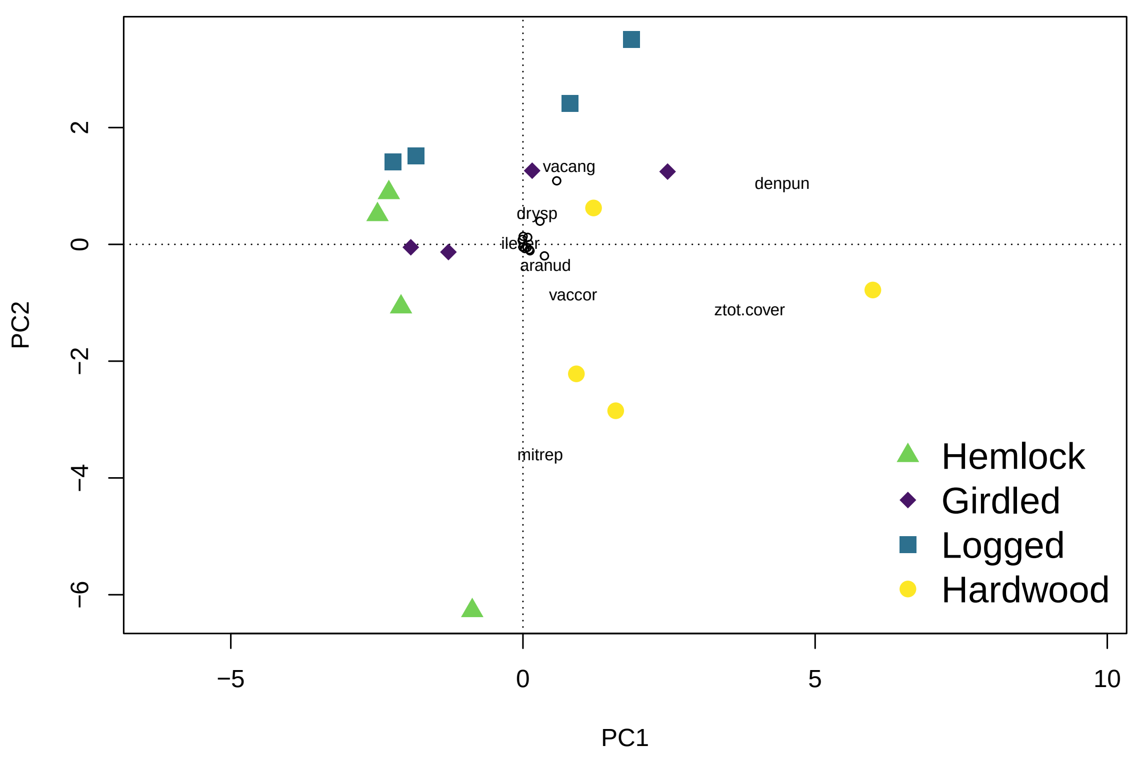
```r
herb.pca <- rda(herb3[,5:23], scale = FALSE)
# if RDA function is used without explanatory variables, then it calculates a PCA
summary(herb.pca)
```

```
Call:
rda(X = herb3[, 5:23], scale = FALSE)

Partitioning of variance:
              Inertia Proportion
Total             439          1
Unconstrained     439          1

Eigenvalues, and their contribution to the variance

Importance of components:
                           PC1      PC2     PC3      PC4      PC5      PC6
Eigenvalue            200.1335 106.6731 47.1189 40.23343 37.52839 3.287174
Proportion Explained    0.4559   0.2430  0.1073  0.09164  0.08548 0.007487
Cumulative Proportion   0.4559   0.6988  0.8062  0.89779  0.98327 0.990759
                          PC7      PC8      PC9      PC10      PC11      PC12
Eigenvalue            1.57620 0.875530 0.599003 0.4158937 0.3433982 0.1054584
Proportion Explained  0.00359 0.001994 0.001364 0.0009473 0.0007822 0.0002402
Cumulative Proportion 0.99435 0.996343 0.997708 0.9986551 0.9994373 0.9996775
                           PC13      PC14      PC15
Eigenvalue            0.0853128 0.0551803 1.093e-03
Proportion Explained  0.0001943 0.0001257 2.490e-06
Cumulative Proportion 0.9998718 0.9999975 1.000e+00
```

The summary output of the PCA is lengthy. It provides eigenvalues and the proportion of variance explained for each axis as well as species and site scores for the first six PCA axes.

Although many PCA axes are generated, the focus is usually on the first two or three axes that explain the most variation in the data. In this case, axis 1 and 2 explain 69% of the variation in the understory plant data, with axis 1 explaining the majority (45%).

```
ordiplot (herb.pca, display = 'sites', type = 'n')
points(herb.pca, dis = "sites", select = which(herb3$trt=="hemlock"), pch = 17,
       cex = 1.5, col = "#73D055FF")
points(herb.pca, dis = "sites", select = which(herb3$trt=="girdled"), pch = 18,
       cex = 1.5, col = "#481567FF")
points(herb.pca, dis = "sites", select = which(herb3$trt=="logged"), pch = 15,
       cex = 1.5, col = "#2D708EFF")
points(herb.pca, dis = "sites", select = which(herb3$trt=="hardwood"), pch = 16,
       cex = 1.5, col = "#FDE725FF")
orditorp(herb.pca, display = 'sp')
legend("bottomright", legend = c("Hemlock", "Girdled", "Logged", "Hardwood"),
       pch = c(17, 18, 15, 16), cex = 1.5, bty = "n",
       col = c("#73D055FF", "#481567FF", "#2D708EFF", "#FDE725FF"))
```



The majority of the variance is explained by PCA axis 1. Based on how the sites fall along axis 1, we can see that it ranges from hemlock dominated forests to hardwood dominated forests. This suggests that there is a change in the understory plant community along this forest gradient.

Direct gradient analysis

Direct gradient analysis (aka constrained ordination) utilizes environmental data in addition to a species by site matrix. When using a direct gradient analysis, the approach assesses whether species composition is related to the measured environmental factors. Therefore, these methods can be used to test hypotheses, which are carried out using permutation procedures. Two commonly used constrained ordination methods are Redundancy Analysis (RDA) and Canonical Correspondence Analysis (CCA).

Redundancy analysis (RDA)

Redundancy analysis combines multiple regression and principal component analysis (PCA) to model multivariate response variables as a function of a set of environmental predictors. RDA is most useful when environmental gradients are short because it assumes linear relationships among species and measured environmental gradients.

This method applies multiple regression of species abundance data as a function of the environmental variables to produce two new matrices: one with predicted values and one with residuals. Constrained and unconstrained ordination axes are extracted from these new matrices using PCA. These axes are linear and orthogonal combinations of the environmental predictor variables that best explain the variation in the species matrix.

We can run an RDA using the rda() function in the package `vegan`. To differentiate this analysis from PCA, we will specify our environmental data set of understory plant species.

```
ants.rda <- rda(ants3[,6:38] ~ ., herb3[,5:23])
summary(ants.rda)
```

```
Call:
rda(formula = ants3[, 6:38] ~ aranud + carpen + denobs + denpun +      drysp + hupluc + ileve

Partitioning of variance:
             Inertia Proportion
Total           1862          1
Constrained     1862          1
Unconstrained      0          0

Eigenvalues, and their contribution to the variance

Importance of components:
                          RDA1     RDA2     RDA3     RDA4     RDA5     RDA6
```

```
Eigenvalue            861.2686 635.5632 211.3264 64.71583 53.89369 19.03651
Proportion Explained    0.4625   0.3413   0.1135  0.03475  0.02894  0.01022
Cumulative Proportion   0.4625   0.8037   0.9172  0.95194  0.98088  0.99110
                          RDA7    RDA8     RDA9     RDA10     RDA11     RDA12
Eigenvalue            8.253510 3.09128 1.931927 1.3615812 0.8908122 0.5335600
Proportion Explained  0.004432 0.00166 0.001037 0.0007311 0.0004783 0.0002865
Cumulative Proportion 0.995536 0.99720 0.998234 0.9989647 0.9994430 0.9997295
                          RDA13     RDA14     RDA15
Eigenvalue            0.3455534 1.163e-01 4.201e-02
Proportion Explained  0.0001855 6.242e-05 2.256e-05
Cumulative Proportion 0.9999150 1.000e+00 1.000e+00

Accumulated constrained eigenvalues
Importance of components:
                          RDA1     RDA2     RDA3     RDA4     RDA5     RDA6
Eigenvalue            861.2686 635.5632 211.3264 64.71583 53.89369 19.03651
Proportion Explained    0.4625   0.3413   0.1135  0.03475  0.02894  0.01022
Cumulative Proportion   0.4625   0.8037   0.9172  0.95194  0.98088  0.99110
                          RDA7    RDA8     RDA9     RDA10     RDA11     RDA12
Eigenvalue            8.253510 3.09128 1.931927 1.3615812 0.8908122 0.5335600
Proportion Explained  0.004432 0.00166 0.001037 0.0007311 0.0004783 0.0002865
Cumulative Proportion 0.995536 0.99720 0.998234 0.9989647 0.9994430 0.9997295
                          RDA13     RDA14     RDA15
Eigenvalue            0.3455534 1.163e-01 4.201e-02
Proportion Explained  0.0001855 6.242e-05 2.256e-05
Cumulative Proportion 0.9999150 1.000e+00 1.000e+00
```

Again, the output is very lengthy. It contains information about the partitioned variance among the constrained and unconstrained axes, as well as the total inertia (i.e., total variance explained by the model, both constrained and unconstrained axes combined). Similar to PCA, it shows the eigenvalues and proportion of variance explained by each RDA axis. Again, although there are many RDA axes, we are only concerned with the first two or three axes that explain the most variation in the ant species data. We can see that the first two axes explain 80% of the variance (46.2% for axis 1 and 34.1% for axis 2).

The importance of the environmental variables can also be assessed. The ordistep() function can be used to build constrained ordination models from the RDA using a stepwise model selection procedure. Significance is determined using permutation tests. It identifies the best or minimum adequate model. This method is one way to test hypotheses with RDA.

First, a null model needs to be created without any environmental predictor variables (mod0). Then, we specify that the scope should be the RDA model with all the environmental predictors. The permutation procedure creates a reduced model with only those environmental variables determined to be important for the community matrix.

```r
mod0.rda <- rda(ants3[,6:38] ~ 1, herb3[,5:23])
ants.rda.red <- ordistep(mod0.rda, scope = formula(ants.rda), direction = "both",
                         permutations = how(nperm = 199))
```

```
Start: ants3[, 6:38] ~ 1

             Df    AIC      F Pr(>F)
+ medvir      1 116.57 7.5117  0.005 **
+ maican      1 117.50 6.2888  0.005 **
+ ztot.cover  1 117.82 5.8941  0.005 **
+ vaccor      1 118.24 5.3759  0.005 **
+ aranud      1 118.29 5.3211  0.010 **
+ denobs      1 119.68 3.7087  0.025 *
+ uvuses      1 119.66 3.7284  0.050 *
+ denpun      1 120.72 2.5932  0.060 .
+ drysp       1 120.89 2.4146  0.145
+ ruball      1 120.94 2.3650  0.145
+ rubida      1 121.86 1.4495  0.235
+ lysqua      1 122.39 0.9475  0.430
+ lysbor      1 122.32 1.0180  0.440
+ ilever      1 122.83 0.5433  0.635
+ vacang      1 123.06 0.3347  0.765
+ rubhis      1 123.08 0.3190  0.860
+ mitrep      1 123.20 0.2124  0.915
+ carpen      1 123.22 0.1900  0.945
+ hupluc      1 123.20 0.2144  0.960
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step: ants3[, 6:38] ~ medvir

         Df    AIC      F Pr(>F)
- medvir  1 121.44 7.5117  0.005 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

             Df    AIC      F Pr(>F)
+ maican      1 111.90 6.7256  0.005 **
+ ztot.cover  1 112.23 6.3213  0.005 **
+ aranud      1 114.44 3.8228  0.015 *
+ denpun      1 114.27 4.0035  0.025 *
```

```
+ rubida      1 115.98 2.2790  0.045 *
+ drysp       1 114.91 3.3410  0.060 .
+ denobs      1 115.85 2.4096  0.075 .
+ vaccor      1 115.25 2.9915  0.090 .
+ ruball      1 115.14 3.1087  0.105
+ uvuses      1 117.30 1.0708  0.380
+ ilever      1 117.49 0.9076  0.440
+ lysbor      1 117.64 0.7782  0.545
+ rubhis      1 117.81 0.6297  0.575
+ lysqua      1 117.87 0.5777  0.655
+ mitrep      1 118.29 0.2312  0.890
+ hupluc      1 118.24 0.2668  0.900
+ carpen      1 118.27 0.2453  0.930
+ vacang      1 118.49 0.0641  0.995
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step: ants3[, 6:38] ~ medvir + maican

         Df    AIC      F Pr(>F)
- maican  1 116.57 6.7256  0.005 **
- medvir  1 117.50 7.9146  0.005 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

             Df    AIC      F Pr(>F)
+ drysp       1 107.25 6.1794  0.005 **
+ ruball      1 108.17 5.1658  0.035 *
+ rubida      1 110.59 2.7581  0.065 .
+ ztot.cover  1 110.90 2.4685  0.070 .
+ ilever      1 111.89 1.6076  0.160
+ uvuses      1 112.01 1.5015  0.205
+ denpun      1 112.04 1.4768  0.215
+ aranud      1 112.29 1.2665  0.255
+ lysbor      1 112.36 1.2102  0.295
+ rubhis      1 112.52 1.0804  0.355
+ vaccor      1 112.64 0.9772  0.420
+ denobs      1 112.81 0.8455  0.475
+ hupluc      1 113.14 0.5794  0.640
+ lysqua      1 113.55 0.2632  0.900
+ mitrep      1 113.72 0.1375  0.925
+ carpen      1 113.62 0.2121  0.960
+ vacang      1 113.79 0.0833  0.980
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step: ants3[, 6:38] ~ medvir + maican + drysp

         Df    AIC      F Pr(>F)
- drysp   1 111.90 6.1794  0.005 **
- medvir  1 114.35 9.1951  0.005 **
- maican  1 114.91 9.9449  0.005 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

             Df    AIC      F Pr(>F)
+ uvuses      1 106.87 1.7681  0.155
+ denpun      1 107.50 1.2714  0.265
+ rubida      1 107.39 1.3524  0.290
+ ztot.cover  1 107.69 1.1255  0.310
+ ilever      1 107.60 1.1965  0.345
+ vaccor      1 107.75 1.0791  0.385
+ hupluc      1 108.11 0.8120  0.480
+ aranud      1 108.24 0.7167  0.555
+ ruball      1 108.38 0.6146  0.625
+ mitrep      1 108.79 0.3192  0.740
+ carpen      1 108.71 0.3770  0.845
+ denobs      1 108.78 0.3296  0.870
+ lysbor      1 108.92 0.2265  0.880
+ rubhis      1 109.00 0.1756  0.910
+ lysqua      1 108.90 0.2454  0.930
+ vacang      1 109.14 0.0727  0.990
```

```r
summary(ants.rda.red) # which environmental variables are important
```

```
Call:
rda(formula = ants3[, 6:38] ~ medvir + maican + drysp, data = herb3[,      5:23])

Partitioning of variance:
              Inertia Proportion
Total          1862.4     1.0000
Constrained    1335.1     0.7169
Unconstrained   527.3     0.2831
```

```
Eigenvalues, and their contribution to the variance

Importance of components:
                          RDA1     RDA2      RDA3      PC1      PC2      PC3
Eigenvalue            745.2449 410.4100 179.44654 272.4958 134.97070 50.62164
Proportion Explained    0.4002   0.2204   0.09635   0.1463   0.07247  0.02718
Cumulative Proportion   0.4002   0.6205   0.71688   0.8632   0.93567  0.96285
                           PC4      PC5       PC6      PC7      PC8       PC9
Eigenvalue            28.38832 19.02878 12.538101 4.575344 2.083680 1.7130500
Proportion Explained   0.01524  0.01022  0.006732 0.002457 0.001119 0.0009198
Cumulative Proportion  0.97810  0.98831  0.995046 0.997503 0.998622 0.9995415
                           PC10    PC11      PC12
Eigenvalue            0.4667099 0.22354 1.637e-01
Proportion Explained  0.0002506 0.00012 8.791e-05
Cumulative Proportion 0.9997921 0.99991 1.000e+00

Accumulated constrained eigenvalues
Importance of components:
                          RDA1     RDA2     RDA3
Eigenvalue            745.2449 410.4100 179.4465
Proportion Explained    0.5582   0.3074   0.1344
Cumulative Proportion   0.5582   0.8656   1.0000
```

Permutation tests also can be used to test the overall fit of the reduced model. The anova() function can assess the importance of the RDA axes for explaining the community data.

```
anova(ants.rda.red, by = 'axis') # which axes are important
```

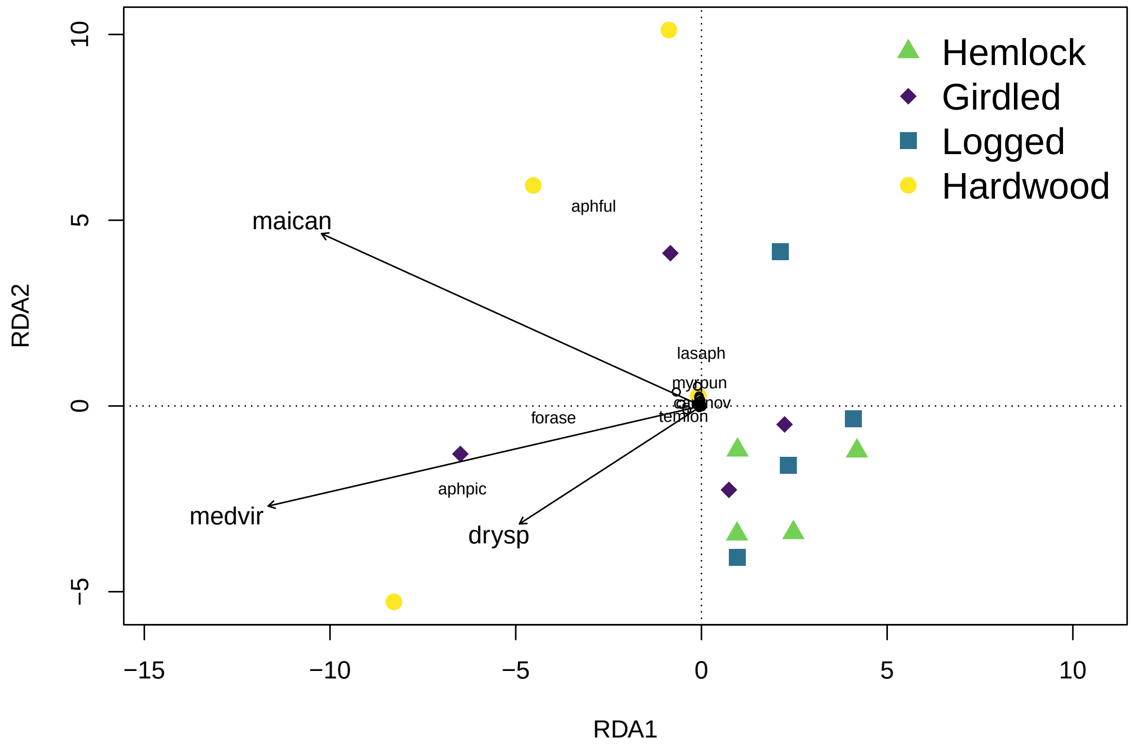
```
Permutation test for rda under reduced model
Forward tests for axes
Permutation: free
Number of permutations: 999

Model: rda(formula = ants3[, 6:38] ~ medvir + maican + drysp, data = herb3[, 5:23])
         Df Variance       F Pr(>F)
RDA1      1   745.24 16.9609  0.001 ***
RDA2      1   410.41  9.3404  0.002 **
RDA3      1   179.45  4.0840  0.008 **
Residual 12   527.27
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

RDA axis 1, axis 2, and axis 3 of the reduced model were found to be important. The reduced model explains less variation in ant species compared to the full RDA model. Axis 1 explains 40% of the variance and axis 2 explains 22%. The reduced model identified three environmental variables to be important for explaining ant communities - the percentage cover of medvir, maican, and drysp. Medvir is the species code for *Medeola virginiana*, a plant species in the lily family known commonly as Indian cucumber. Maican is the species code for *Maianthemum canadense*, a perennial flowering plant native to the northeastern US and Canada commonly known as Canada mayflower. Drysp is the species code for *Dryopteris carthusiana*, a native fern species commonly known as spinulose woodfern.

We can plot the reduced model with sites categorized by the hemlock treatment to describe the overall patterns in ant species and understory plant data. Before we plot, we need to create a vector that contains the environmental variables that were determined to be important in the model. We can use the `envfit` function to calculate the correlation of important environmental variables with the ordination axes, which allows us to add them as vectors to the plot.

```
herb_vectors_rda <- envfit(ants.rda.red, herb3[, c(9,14,15)])
```

```
ordiplot(ants.rda.red, display= c('si', 'cn'), type = 'n')
points(ants.rda.red, dis = "sites", select = which(herb3$trt=="hemlock"),
       pch = 17, cex = 1.5, col = "#73D055FF")
points(ants.rda.red, dis = "sites", select = which(herb3$trt=="girdled"),
       pch = 18, cex = 1.5, col = "#481567FF")
points(ants.rda.red, dis = "sites", select = which(herb3$trt=="logged"),
       pch = 15, cex = 1.5, col = "#2D708EFF")
points(ants.rda.red, dis = "sites", select = which(herb3$trt=="hardwood"),
       pch = 16, cex = 1.5, col = "#FDE725FF")
orditorp(ants.rda.red, display = 'sp')
plot(herb_vectors_rda, add = TRUE, col = "black", cex = 1)
legend("topright", legend = c("Hemlock", "Girdled", "Logged", "Hardwood"),
       pch = c(17, 18, 15, 16), cex = 1.5, bty = "n",
       col = c("#73D055FF", "#481567FF", "#2D708EFF", "#FDE725FF"))
```



Canonical correspondence analysis (CCA)

Canonical correspondence analysis is similar to RDA, but there are a couple differences. CCA combines multiple regression and correspondence analysis (CA) to model multivariate response variables as a function of a set of environmental predictors. CCA is most useful when environmental gradients are long because it assumes unimodal relationships among species and measured environmental gradients.

We can run a CCA using the cca() function in the package `vegan`.

```
ants.cca <- cca(ants3[,6:38] ~ ., herb3[,5:23])
summary(ants.cca)
```

```
Call:
cca(formula = ants3[, 6:38] ~ aranud + carpen + denobs + denpun +      drysp + hupluc + ileve
```

```
Partitioning of scaled Chi-square:
              Inertia Proportion
Total           1.413          1
Constrained     1.413          1
Unconstrained   0.000          0

Eigenvalues, and their contribution to the scaled Chi-square

Importance of components:
                        CCA1   CCA2   CCA3    CCA4    CCA5    CCA6    CCA7
Eigenvalue            0.3884 0.2157 0.2024 0.13699 0.10486 0.08767 0.06955
Proportion Explained  0.2750 0.1527 0.1433 0.09697 0.07423 0.06207 0.04923
Cumulative Proportion 0.2750 0.4277 0.5710 0.66795 0.74218 0.80425 0.85348
                         CCA8    CCA9   CCA10   CCA11   CCA12    CCA13    CCA14
Eigenvalue            0.06501 0.05056 0.03965 0.02460 0.01647 0.005975 0.003515
Proportion Explained  0.04602 0.03579 0.02807 0.01741 0.01166 0.004230 0.002488
Cumulative Proportion 0.89951 0.93530 0.96337 0.98078 0.99244 0.996669 0.999157
                          CCA15
Eigenvalue            0.0011906
Proportion Explained  0.0008428
Cumulative Proportion 1.0000000

Accumulated constrained eigenvalues
Importance of components:
                        CCA1   CCA2   CCA3    CCA4    CCA5    CCA6    CCA7
Eigenvalue            0.3884 0.2157 0.2024 0.13699 0.10486 0.08767 0.06955
Proportion Explained  0.2750 0.1527 0.1433 0.09697 0.07423 0.06207 0.04923
Cumulative Proportion 0.2750 0.4277 0.5710 0.66795 0.74218 0.80425 0.85348
                         CCA8    CCA9   CCA10   CCA11   CCA12    CCA13    CCA14
Eigenvalue            0.06501 0.05056 0.03965 0.02460 0.01647 0.005975 0.003515
Proportion Explained  0.04602 0.03579 0.02807 0.01741 0.01166 0.004230 0.002488
Cumulative Proportion 0.89951 0.93530 0.96337 0.98078 0.99244 0.996669 0.999157
                          CCA15
Eigenvalue            0.0011906
Proportion Explained  0.0008428
Cumulative Proportion 1.0000000
```

As you can see, the output looks very similar to RDA, and you can interpret it the same way. We can see that the first two axes explain 42% of the variance (27.5% for axis 1 and 15.2% for axis 2). We can use the same permutation procedures to assess the importance of the environmental variables via a reduced model as well as the importance of the axes.

```r
mod0.cca <- cca(ants3[,6:38] ~ 1, herb3[,5:23])
ants.cca.red <- ordistep(mod0.cca, scope = formula(ants.cca), direction = "both",
                          permutations = how(nperm = 199))
```

```
Start: ants3[, 6:38] ~ 1

             Df    AIC      F Pr(>F)
+ medvir      1 78.899 2.6219  0.025 *
+ uvuses      1 78.960 2.5580  0.025 *
+ aranud      1 79.260 2.2504  0.055 .
+ denpun      1 79.516 1.9933  0.095 .
+ maican      1 79.623 1.8864  0.115
+ denobs      1 79.510 1.9991  0.140
+ rubida      1 79.799 1.7128  0.150
+ drysp       1 79.757 1.7535  0.200
+ ruball      1 79.975 1.5405  0.240
+ vaccor      1 80.019 1.4983  0.285
+ ilever      1 80.352 1.1791  0.420
+ rubhis      1 80.423 1.1117  0.460
+ carpen      1 80.431 1.1044  0.565
+ ztot.cover  1 80.381 1.1509  0.605
+ lysqua      1 80.850 0.7139  0.865
+ vacang      1 81.249 0.3513  0.875
+ lysbor      1 80.931 0.6397  0.920
+ mitrep      1 81.056 0.5251  0.945
+ hupluc      1 80.915 0.6536  0.950
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step: ants3[, 6:38] ~ medvir

         Df    AIC      F Pr(>F)
- medvir  1 79.645 2.6219  0.015 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

             Df    AIC      F Pr(>F)
+ denobs      1 78.410 2.1876  0.065 .
+ maican      1 78.693 1.9213  0.080 .
+ rubida      1 78.680 1.9340  0.095 .
+ denpun      1 78.613 1.9968  0.105
```

```
+ drysp       1 78.643 1.9679  0.105
+ ruball      1 78.891 1.7382  0.175
+ ztot.cover  1 79.090 1.5559  0.230
+ aranud      1 79.233 1.4261  0.325
+ uvuses      1 79.447 1.2346  0.355
+ rubhis      1 79.526 1.1646  0.435
+ ilever      1 79.494 1.1925  0.470
+ carpen      1 79.545 1.1475  0.540
+ hupluc      1 79.835 0.8933  0.705
+ vaccor      1 80.141 0.6309  0.810
+ lysbor      1 80.070 0.6912  0.885
+ lysqua      1 80.094 0.6709  0.900
+ mitrep      1 80.275 0.5169  0.915
+ vacang      1 80.633 0.2175  0.975
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(ants.cca.red) # which environmental variables are important
```

```
Call:
cca(formula = ants3[, 6:38] ~ medvir, data = herb3[, 5:23])

Partitioning of scaled Chi-square:
              Inertia Proportion
Total          1.4126     1.0000
Constrained    0.2228     0.1577
Unconstrained  1.1898     0.8423

Eigenvalues, and their contribution to the scaled Chi-square

Importance of components:
                        CCA1    CA1    CA2    CA3     CA4     CA5     CA6
Eigenvalue            0.2228 0.3382 0.2031 0.1523 0.11121 0.09271 0.07178
Proportion Explained  0.1577 0.2394 0.1438 0.1078 0.07873 0.06563 0.05081
Cumulative Proportion 0.1577 0.3972 0.5410 0.6488 0.72752 0.79316 0.84397
                          CA7     CA8     CA9    CA10    CA11     CA12     CA13
Eigenvalue            0.06926 0.05056 0.04142 0.02604 0.02207 0.005977 0.003594
Proportion Explained  0.04903 0.03579 0.02932 0.01843 0.01563 0.004231 0.002544
Cumulative Proportion 0.89300 0.92880 0.95812 0.97655 0.99218 0.996410 0.998954
                          CA14
Eigenvalue            0.001478
```

```
Proportion Explained  0.001046
Cumulative Proportion 1.000000

Accumulated constrained eigenvalues
Importance of components:
                        CCA1
Eigenvalue            0.2228
Proportion Explained  1.0000
Cumulative Proportion 1.0000
```

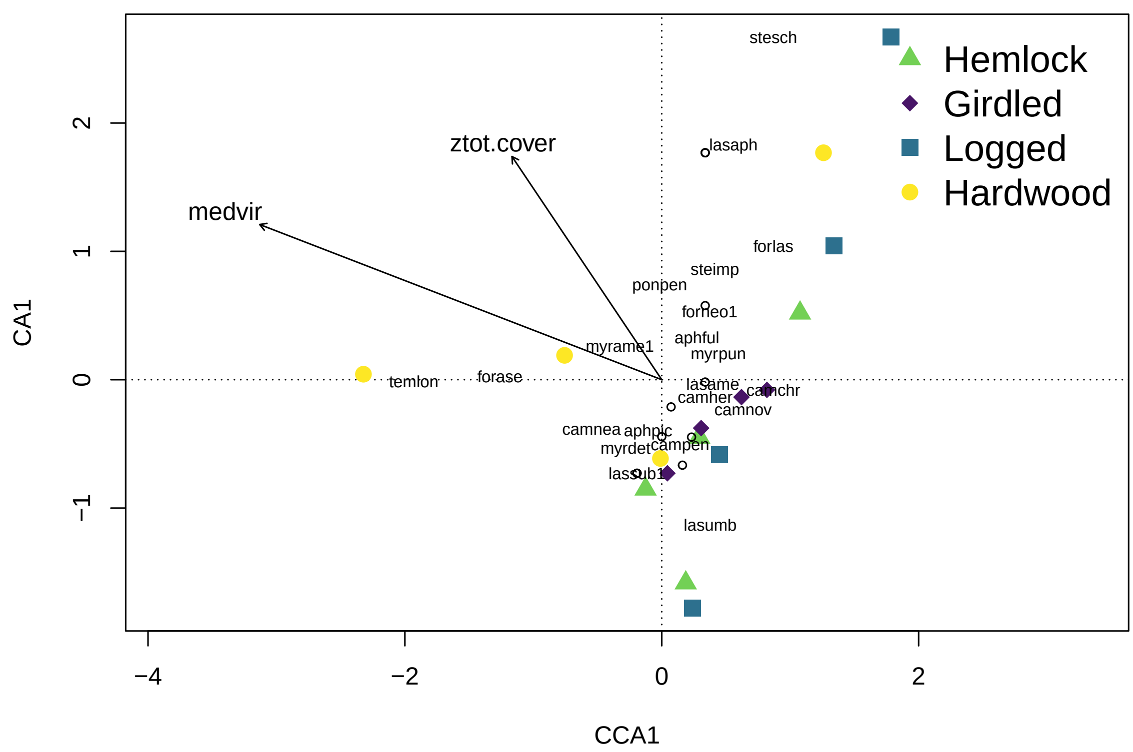
```r
anova(ants.cca.red, by = 'axis') # which axes are important
```

```
Permutation test for cca under reduced model
Forward tests for axes
Permutation: free
Number of permutations: 999

Model: cca(formula = ants3[, 6:38] ~ medvir, data = herb3[, 5:23])
         Df ChiSquare      F Pr(>F)
CCA1      1   0.22282 2.6219   0.02 *
Residual 14   1.18976
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

CCA axis 1 of the reduced model was found to be important. The reduced CCA model explained less variation in ant species compared to the full CCA model. Axis 1 explains 15.7% of the variance. The reduced model identified one environmental variable as important for explaining ant communities - percentage cover of medvir. As in the RDA, medvir is the species code for *Medeola virginiana*, a plant species in the lily family known commonly as Indian cucumber.

We can plot the reduced model with sites categorized by the hemlock treatment to describe the overall patterns in ant species and understory plant data. I've also included total percentage cover of understory plants (ztot.cover) in the figure. It is important to remember that your expertise of the system is also important for understanding and interpreting the output of these analyses. If there is an environmental variable that does not show up in the model selection procedure, but you, as the researcher, feel is important, you can always include it in the figure. Just be sure to indicate that it was not determined to be an important driver of ant communities using these methods.

```
herb_vectors_cca <- envfit(ants.cca.red, herb3[, c(15, 23)])

ordiplot(ants.cca.red, display = c('si', 'cn'), type = 'n')
points(ants.cca.red, dis = "sites", select = which(herb3$trt=="hemlock"),
       pch = 17, cex = 1.5, col = "#73D055FF")
points(ants.cca.red, dis = "sites", select = which(herb3$trt=="girdled"),
       pch = 18, cex = 1.5, col = "#481567FF")
points(ants.cca.red, dis = "sites", select = which(herb3$trt=="logged"),
       pch = 15, cex = 1.5, col = "#2D708EFF")
points(ants.cca.red, dis = "sites", select = which(herb3$trt=="hardwood"),
       pch = 16, cex = 1.5, col = "#FDE725FF")
orditorp(ants.cca.red, display = 'sp')
plot(herb_vectors_cca, add = TRUE, col = "black", cex = 1)
legend("topright", legend = c("Hemlock", "Girdled", "Logged", "Hardwood"),
       pch = c(17, 18, 15, 16), cex = 1.5, bty = "n",
       col = c("#73D055FF", "#481567FF", "#2D708EFF", "#FDE725FF"))
```



How to select which method (CCA or RDA) to use

The most appropriate model to use depends on the length of the environmental gradients represented in the study. CCA assumes that the environmental variables represent a unimodal distribution and that sites sampled are representative of both ends of this gradient. RDA assumes a linear distribution such that sites sampled represent only a portion of the unimodal distribution. Generally, studies that implement experimental treatments and/or are smaller in scale will have linear environmental gradients. This is because smaller scale studies usually do not incorporate the full range of environmental variation experienced by species in a community to produce a unimodal response.

The length of the gradient can be estimated based on the data using a Detrended Correspondence Analysis (DCA). DCA is an indirect gradient method that was originally developed to remove the arch effect from Correspondence Analysis (CA). This method is no longer used for community analysis except for the purpose of identifying gradient lengths. DCA repeatedly splits the main axes into segments until the best solution is found. The length of the first axis can be used to determine whether an RDA (linear, axis length < 4) or CCA (unimodal, axis length > 4) model is best for the data set.

```
DCA <- decorana(ants3[,6:38])
DCA
```

```
Call:
decorana(veg = ants3[, 6:38])

Detrended correspondence analysis with 26 segments.
Rescaling of axes with 4 iterations.
Total inertia (scaled Chi-square): 1.4126

                       DCA1   DCA2    DCA3    DCA4
Eigenvalues          0.3548 0.1835 0.14205 0.09275
Additive Eigenvalues 0.3548 0.1828 0.13337 0.05265
Decorana values      0.3884 0.1422 0.05087 0.02241
Axis lengths         2.1980 1.2314 1.11022 1.26230
```

The length of DCA axis 1 is is 2.1 (less than 4), so RDA is the more appropriate analysis for the ant data set.

R Activity

1. We will use another open source data set from the NSF Harvard Forest Long-term Ecological Research (LTER) site. These data are spiders collected in the Hemlock Removal Experiment. Remember, this experiment includes four treatments (Hemlock girdled, Hemlock logged, Hemlock control, and Hardwood control) each replicated across two ($n = 2$) 90 x 90 m plots. Load the `HarvardForest_spiders` data into R. We will characterize spider communities among these four treatments. Load the `HarvardForest_HerbLayer` data into R. We will also assess the relationship among spiders and understory plants.

2. Before running any analyses, you will have to do some data wrangling. First, create a new variable `abundance` by summing the counts of adult male and female spiders. Next, change the data set from `long` format to `wide` format using spider genus as the taxonomic resolution (i.e., each column should be a spider genus). Be sure that the new data frame includes the predictor variables `block`, `plot`, `replicate`, `treatment`, and `sampling method`. Then, remove any columns that do not have count data (some genera are indicated as immatures with imm), as well as three columns with unidentified spiders (LinytoID, LinyToID, and unk_toID). Change the variables `treatment` and `sampling method` to factors. Provide a summary of the data set.

3. Calculate a dissimilarity matrix for the spider data using the bray-curtis method. Run a nonmetric multidimensional scaling (NMDS) model using the metaMDS() function on the spider genera data. Provide a figure showing the treatment groups with 95% confidence interval ellipses. Report the model stress.

4. Run a permutational multivariate analysis of variance (PERMANOVA) using adonis2(). Model the bray-curtis distance matrix as a function of `treatment`. Provide the PERMANOVA output.

5. Interpret the output (using the figure and PERMANOVA results).

6. Now, some data wrangling for the understory herb data. Change the data set from `long` format to `wide` format using species as the taxonomic resolution (i.e., each column should be a plant species). Be sure that the new data frame includes the predictor variables `year`, `block`, `trt`, `plot`, and `subplot`. Then, subset the data by year 2008, so it aligns with the spider data. Lastly, double check that the number of samples per plot aligns between the spider and plant data sets. Each plot should have 10 samples (indicated as replicates or subplots).

7. Run a detrended correspondence analysis (DCA) to determine the most appropriate constrained model for these data. Provide the output.

8. Run the appropriate model (RDA or CCA). Provide a figure.

9. Interpret the results.